The influence matrix (aka Hat matrix, projection matrix) is

$$\underset{=}{H} = \underset{=}{X}(\underset{=}{X}^T\underset{=}{X})^{-1}\underset{=}{X}^T$$

The fitted values are given by $\underset{=}{X}\hat{\underset{=}{\beta}} = \underset{=}{H}\underset{=}{y} = \underset{=}{X}(\underset{=}{X}^T\underset{=}{X})^{-1}\underset{=}{X}^T\underset{=}{y} = \underset{=}{H}\underset{=}{y}$.
This is the projection of $\underset{=}{y}$ onto the space spanned by the columns of $\underset{=}{X}$.

(1) The influence matrix $\underset{=}{H}$ is idempotent, i.e. $\underset{=}{H}^2 = \underset{=}{H}$.

$$\underset{=}{H}^2 = \underset{=}{X}(\underset{=}{X}^T\underset{=}{X})^{-1}\underset{=}{X}^T\,\underset{=}{X}(\underset{=}{X}^T\underset{=}{X})^{-1}\underset{=}{X}^T = \underset{=}{X}(\underset{=}{X}^T\underset{=}{X})^{-1}\underset{=}{X}^T = \underset{=}{H}.$$

(2) The "trace" of a square matrix is the sum of elements on the main diagonal. For the influence matrix

$$tr(\underset{=}{H}) = p. \quad \leftarrow \text{no. of columns of } \underset{=}{X}.$$

(3) H is symmetric.

(4) $\underset{=}{H}\underset{=}{X} = \underset{=}{X}$ since $\underset{=}{H}\underset{=}{X} = \underset{=}{X}(\underset{=}{X}^T\underset{=}{X})^{-1}\underset{=}{X}^T\underset{=}{X} = \underset{=}{X}$.

Fitted values are unbiased for $E(\underset{=}{Y})$.

$$E(\underset{=}{X}\hat{\underset{=}{\beta}}) = \underset{=}{X}\,E(\hat{\underset{=}{\beta}}) = \underset{=}{X}\underset{=}{\beta} = E(\underset{=}{Y})$$

Covariance matrix of the fitted values

$$cov(\underset{=}{X}\hat{\underset{=}{\beta}}) = \underset{=}{X}\,cov(\hat{\underset{=}{\beta}})\,\underset{=}{X}^T = \underset{=}{X}\cdot\sigma^2(\underset{=}{X}^T\underset{=}{X})^{-1}\cdot\underset{=}{X}^T = \sigma^2\underset{=}{H}$$

The residuals are $\hat{\underset{=}{e}} = \underset{=}{y} - \hat{\underset{=}{y}} = \underset{=}{y} - \underset{=}{H}\underset{=}{y} = (\underset{=}{I} - \underset{=}{H})\underset{=}{y}$

The residuals have zero expectation.

$$E\left[(\underset{=}{I} - \underset{=}{H})\underset{=}{Y}\right] = (\underset{=}{I} - \underset{=}{H})E(\underset{=}{Y}) = (\underset{=}{I} - \underset{=}{H})\underset{=}{X}\underset{=}{\beta} = (\underset{=}{X} - \underset{=}{H}\underset{=}{X})\underset{=}{\beta} = 0$$

since $\underset{=}{H}\underset{=}{X} = \underset{=}{X}$.

$$\text{cov}(\hat{e}) = \text{cov}((I-H)Y) = (I-H)\,\text{cov}(Y)\cdot(I-H)^T$$

Since H is symmetric $(I-H)^T = (I-H)$

$$= (I-H)\sigma^2 I\cdot(I-H)$$

$$= \sigma^2(I-H)^2 = \sigma^2(I-2H+H) = \sigma^2(I-H)$$

Standardised residuals : Since the residuals $\hat{e}$ has covariance matrix $\sigma^2(I-H)$, they won't have the same variance. The standardised residuals are

$$r_i = \frac{\hat{e}_i}{\hat{\sigma}_u\sqrt{1-H_{ii}}}$$

Leverage : The diagonal elements of H are called leverages.
∴ A large leverage will make the $\hat{e}_i$ small. so the fit of the model is drawn to observation y_i.

* Note leverage is determined by X not y. Since $\text{tr}(H)=p$ on average leverage is p/n.

Rule of thumb : leverages larger than $2p/n$ should be investigated.

Influential observations : An influential point is one whose removed from the dataset would cause a large change in fit.

Influential observation tend to have large leverage or large residual or both.

Let subscript (i) denote fit without observation i.

- Change in fitted values : $X^*(\hat{\beta}-\hat{\beta}_{(i)}) = \hat{y}-\hat{y}_{(i)}$.

- Change in coefficients : $\hat{\beta} - \hat{\beta}_{(i)}$

- Cook's statistic $D_i = \dfrac{(\hat{y} - \hat{y}_{(i)})^T (\hat{y} - \hat{y}_{(i)})}{p\,\hat{\sigma}^2} = \dfrac{1}{p} r_i^2 \dfrac{H_{ii}}{1 - H_{ii}}$

$D_i \approx 1$ strong indication of substantial influence

Rule of Thumb : $D_i > 4/(n-p)$ worth investigating.